

ME 300 Thermodynamics – Summer 2023

Exam 2

(Part I) _____ (30 pts)

SOLUTIONS

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

Directions

1. Open the online answers portion of the exam and download the exam (this document).
2. Make sure to show all your work!
 - a. *If you are using a tablet*, solve the problems directly on this document, showing all your work and writing as clearly as possible.
 - b. *If you are writing on paper*, solve the problems on a separate piece of paper (you do not have to print the exam), showing all your work and writing as clearly as possible.
3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD – this will be distracting to the other students.
5. Once the answers portion of the exam closes, scan or take very nice pictures of your work and upload it to the upload portion of the exam in a PDF document. I trust that you will not change anything on your exam after the answers portion of the exam closes. I can't enforce this – I'm really just relying on you to do the right thing. Please keep to the honor code.
6. The timeline for the exam will be the following (all times in EDT, Thursday, July 20):
 - a. Exam opens (answers portion and upload portion): 6:00 PM
 - b. Answers portion of the exam closes: 7:30 PM
 - c. Upload portion of the exam closes: 7:45 PM
7. You can anonymously report any incidents of cheating using the following Google form:
<https://forms.gle/E4wFjDtvNuginVkp9>

Academic Integrity Pledge

I pledge that I am adhering to Penn State's Code of Conduct when taking this exam.

- I will not communicate with anyone during the exam except Dr. O'Connor.
- I will use only the allowed materials (calculator, notes, tables, and book), I will only access the internet to download the exam, view the e-book and notes on Canvas (no videos), and to submit my answers and upload the exam. I will not use any generative AI (ChatGPT) during the exam.
- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers.
In the online answers, please type your full legal name to agree to the academic integrity pledge.

Part I: Concept Questions

1. (4 points) Can the thermal efficiency of a heat engine ever reach one? *Answer the multiple-choice question in Canvas.*

YES

NO +1

Explain your response to this question. *Type your answer in the box in Canvas.*

The Kelvin-Planck Statement⁺¹ of the 2nd law says that heat cannot be continuously and completely converted to work. Since $\eta_{th} = \frac{\bar{W}_{net}}{Q_{in}}$, and $Q_{in} > \bar{W}_{net}$, then $\eta_{th} < 1$. +1

2. (16 points) Identify the correct sign of the change in entropy for the following four processes.

- a) An ideal gas undergoes an isothermal increase in volume. Is the change in entropy of the gas is: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO +2

LESS THAN ZERO

EQUAL TO ZERO

$$\Delta S = C_V \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{V_2}{V_1} \right) +1$$

$T_2 = T_1 \rightarrow 0$ $V_2 > V_1 \rightarrow 0$

- b) An ideal gas undergoes isobaric temperature decrease. Is the change in entropy of the gas is: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO

LESS THAN ZERO +2

EQUAL TO ZERO

$$\Delta S = C_P \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right) +1$$

$T_2 < T_1 \rightarrow 0$ $P_2 = P_1 \rightarrow 0$

- c) Identify the equations you used to explain the answers to these previous two questions and what assumptions are necessary to use these equations. *Type your answer in the box in Canvas.*

→ see above

→ ideal gas +1

$C_P, C_V = \text{const}$ +1

- d) A solid block is heated reversibly. Is the change in entropy of the block: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO

+2

LESS THAN ZERO

EQUAL TO ZERO

- e) Water is adiabatically compressed in an irreversible process. Is the change in entropy of the water: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO

+2

LESS THAN ZERO

EQUAL TO ZERO

- f) Identify the equation you used to explain the answers to these previous two questions and what the terms mean in that equation. *Type your answer in the box in Canvas.*

$$\Delta S = \int \frac{\delta Q}{T} + \Phi$$

+2

change in
entropy due
to heat
transfer

entropy generation from irreversibilities

+1

+1

3. (10 points) A ^{steady flow} compressor is used to increase the pressure of a fluid and it has an isentropic efficiency of 0.85.

- a) What is the sign of the power necessary to run the compressor? *Answer the multiple-choice question in Canvas.*

POSITIVE

NEGATIVE

+2

EQUAL TO ZERO

- b) Explain your response to the previous question – use an equation to support your answer. *Type your answer in the box in Canvas.*

$$\text{compressor: } \dot{m} (\Delta h + \cancel{\Delta \frac{1}{2} v^2} + \cancel{\Delta p v}) = \cancel{\dot{Q}} - \dot{W}$$

$$\dot{W} = -\dot{m} (h_2 - h_1) \rightarrow h_2 > h_1 \text{ since compressor adds energy through work}$$

+2

$$\rightarrow \dot{W} < 0$$

+1

- c) What is the sign of the change in entropy of the fluid as it goes through the compressor? *Answer the multiple-choice question in Canvas.*

POSITIVE

+2

NEGATIVE

EQUAL TO ZERO

- d) Explain your response to the previous question – use an equation to support your answer. *Type your answer in the box in Canvas.*

$\eta_{\text{isen}} < 1$, which means process is irreversible +1

$$\Delta S' = \int \frac{\delta Q}{T} + \Phi$$

+2

adiabatic $\rightarrow \Delta S' > 0$

so since irreversible

Part II: Short Problems

1. (15 points) Helium slowly undergoes a process from $T_1 = 800$ K and $P_1 = 0.1$ MPa to $T_2 = 850$ K and $P_2 = 1$ MPa. Answer the following questions about this process using the following properties for helium: $R = 2077.3$ J/kg-K and $c_p = 5190$ J/kg-K.

- a) (3 points) What assumptions can you make about this fluid to solve for the properties of the fluid at both states? *and proves*
Type your answer in the box in Canvas.
- b) (8 points) What is the change in entropy of the fluid during this process? *mass-specific* *in [J/kg-K]*
Show your work below or on a separate page and enter this value in the Canvas quiz.
- c) (4 points) Can this process be an adiabatic process? Explain why or why not – use an equation to support your answer. *Type your answer in the box in Canvas.*

a) Ideal gas +1
 $c_p = \text{const.}$ +1
 quasi-ef. +1

b) $\Delta S = c_p \ln(T_2/T_1) - R \ln(P_2/P_1)$ +4
 $= 5190 \ln\left(\frac{850}{800}\right) - 2077.3 \ln\left(\frac{1}{0.1}\right)$
 $\Delta S = -4468.5 \text{ J/kg-K}$ +2

c) No. $\Delta S = \int \frac{\delta Q}{T} + \Phi$ +2
 $\underbrace{\int \frac{\delta Q}{T}}_{\text{can be } > 0 \text{ or } < 0}$ $\underbrace{\Phi}_{\text{always } \geq 0}$ +1
 +1

$\Rightarrow \Delta S < 0$ in this problem
 and the only way that can
 happen is through heat transfer.
 +2

2. (25 points) A reversible cycle operates between a maximum temperature of 1800 K and a minimum temperature of 300 K with a working fluid of air ($R=287 \text{ J/kg-K}$, $c_p=1001 \text{ J/kg-K}$, and $\gamma=1.4$). Answer the following questions about the cycle and the processes in the cycle.

- a) (3 points) What is the thermal efficiency of this cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- b) (6 points) If 100 kJ of heat is added to the cycle, how much net work is produced by the cycle in [kJ]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (2 points) What are the assumptions necessary to solve for any of the states in the cycle or the performance of any of its components? *Type your answer in the box in Canvas.*
- d) (7 points) The first step of the cycle is an adiabatic compression from $T_1=300 \text{ K}$ and $P_1 = 0.1 \text{ MPa}$ to a pressure of $P_2=1.2 \text{ MPa}$. What is the temperature of the air after that first process? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- e) (7 points) The second step of the cycle is an isobaric heat addition step in a passive heat exchanger that brings the fluid to its maximum temperature in the cycle. If the mass flow rate of air is 10 kg/s, what is the rate of heat addition in this step? *Show your work below or on a separate page and enter this value in the Canvas quiz.*

Given: $T_{\max} = 1800 \text{ K}$
 $T_{\min} = 300 \text{ K}$

c) Assume: reversible $+1/2$ steady flow $+1/2$
 ideal gas $+1/2$
 $c_p = \text{const}$ $+1/2$

a) Reversible cycle $\rightarrow \eta_{\text{th}} = \eta_{\text{max}} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{1800} \rightarrow \boxed{\eta_{\text{th}} = 0.83}$ $+2$

b) $Q_H = 100 \text{ kJ}$
 $\bar{W}_{\text{net}} = ?$

$\rightarrow \eta_{\text{th}} = \frac{\bar{W}_{\text{net}}}{Q_{\text{in}}} \rightarrow \bar{W}_{\text{net}} = \eta_{\text{th}} Q_{\text{in}} = (0.83)(100)$
 $\boxed{\bar{W}_{\text{net}} = 83 \text{ kJ}}$ $+2$

d) $T_1 = 300 \text{ K}$
 $P_1 = 0.1 \text{ MPa}$
 $P_2 = 1.2 \text{ MPa}$

c) ideal gas
 $c_p = \text{const.}$
 adiabatic
 reversible } isentropic

$\rightarrow T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 300 \left(\frac{1.2}{0.1} \right)^{\frac{0.4}{1.4}}$ $+3$
 $\boxed{T_2 = 610.18 \text{ K}}$ $+2$

d) heat exchanger: $\Delta ke = \Delta pe = 0$ steady flow
 $\dot{W} = 0$

$$\rightarrow \dot{m} (\Delta h + \underbrace{\Delta ke}_{=0} + \underbrace{\Delta pe}_{=0}) = \dot{Q} - \underbrace{\dot{W}}_{=0} \quad +2$$

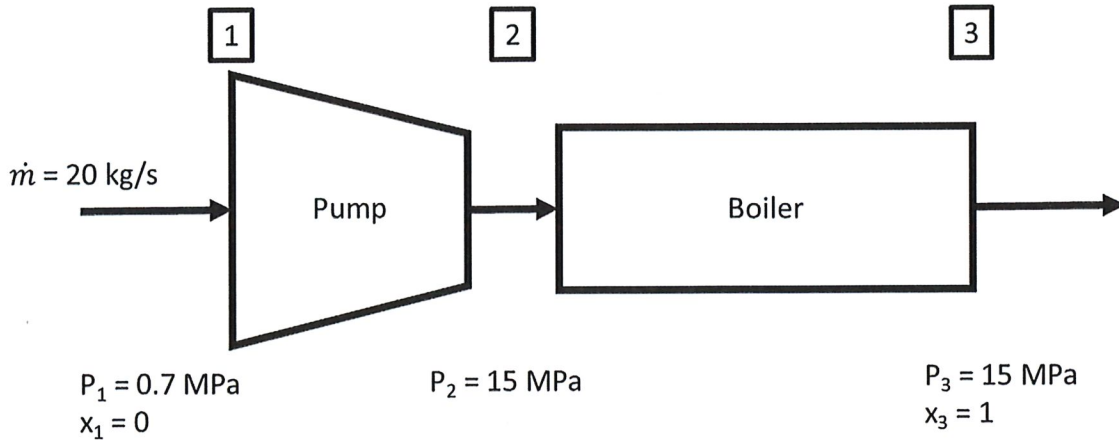
$$\dot{Q} = \dot{m} (h_3 - h_2) \quad +3$$

$$\rightarrow \text{ideal gas, } c_p = \text{const: } \dot{Q} = \dot{m} c_p (T_3 - T_2) \\ = (10)(1001)(1800 - 610.18)$$

$$\boxed{\dot{Q} = 11.91 \text{ MW}} \quad +2$$

Part III: Analysis (30 points)

The story: High-pressure steam is required for gasification, which is a key process in the synthesis of alternative fuels like sustainable aviation fuels. In these systems, water is pumped up to high pressure and then heated to create high-pressure steam. The diagram below shows the two components in this system – assume both components operate reversibly and with a steady flow rate. Identify any tables you use at each step of the problem. If you use a superheated vapor or subcooled liquid table, do not interpolate between values – use the closest value on the table.



- (3 points) List the assumptions necessary to calculate the state points before and after the pump and the heat exchanger, as well as the power and heat flows in this system. *Type your answer in the box in Canvas.*
- (8 points) Calculate the temperature of the water after the pump. Identify the table you used to calculate the state point. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (6 points) Calculate the power necessary to run the pump in [kW]. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (8 points) Calculate the rate at which heat is added to the water in the boiler in [kW]. Identify the table you used to calculate any state points. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (5 points) Draw these processes on a T-s diagram, labeling the state points and phases of fluid at each state. *Draw your diagram below – do not enter anything into the Canvas quiz.*

a) S.C.S. $+1/2$ Pump: $\Delta ke = \Delta pe = 0$ Boiler: $\Delta ke = \Delta pe = 0$
 quasi-ef. $+1/2$ $\dot{Q} = 0$ $+1/2$ $\dot{W} = 0$ $+1/2$
 reversible $+1/2$
 steady flow $+1/2$

b) State 1: $P_1 = 0.7 \text{ MPa}$ } Table D.2 $\rightarrow S_1 = S_f = 1.9918 \text{ kJ/kg-K}$ $+1$
 $x_1 = 0$ $h_1 = h_f = 697.00 \text{ kJ/kg}$ $+1$

State 2: $P_2 = 15 \text{ MPa}$

$$s_2 = s_1 = 1.9918 \text{ kJ/kg} \cdot \text{K} \xrightarrow{+1} \text{Table D.2: } s_2 < s_f \rightarrow \text{liquid} \quad +1$$

$$\xrightarrow{+1} \text{Table D.4: } T_2 \approx 440 \text{ K} \quad +1$$

$$h_2 = 713.44 \text{ kJ/kg}$$

c) steady flow $\rightarrow \dot{m}(\Delta h + \cancel{\Delta \text{ke}} + \cancel{\Delta \text{pe}}) = \dot{Q} - \dot{W} \quad +1$

pump $\rightarrow \dot{m}(h_2 - h_1) = -\dot{W} \quad +2$

$$\dot{W} = -\dot{m}(h_2 - h_1) \quad +1$$

$$= -(20)(713.44 - 697.00) \rightarrow \boxed{\dot{W} = -328.8 \text{ kW}} \quad +1$$

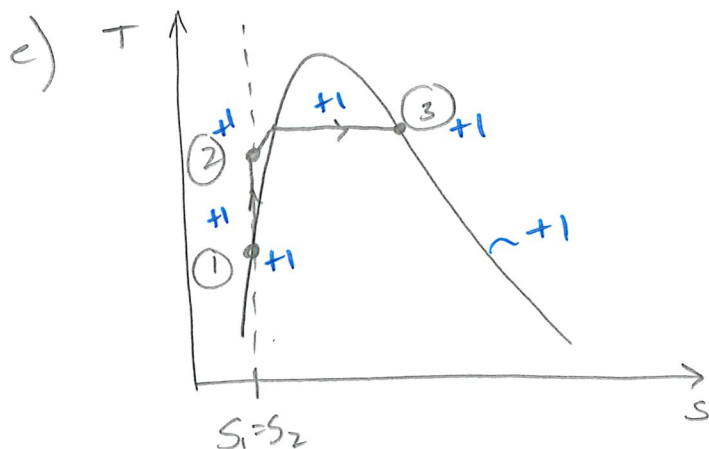
d) State 3: $P_3 = 15 \text{ MPa}$ $\left\{ \begin{array}{l} +1 \\ \text{Table D.2} \end{array} \right. \xrightarrow{+1} h_3 = h_g = 2610.7 \text{ kJ/kg} \quad +1$
 $x_3 = 1$

Steady-flow $\rightarrow \dot{m}(\Delta h + \cancel{\Delta \text{ke}} + \cancel{\Delta \text{pe}}) = \dot{Q} - \cancel{\dot{W}} \quad +1$

boiler $\rightarrow \dot{Q} = \dot{m}(h_3 - h_2) \quad +2$

$$= (20)(2610.7 - 713.44)$$

$$\boxed{\dot{Q} = 37945.2 \text{ kW}} \quad +1$$



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